

MSYNC: pseudosynchrony

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S1 Introduction

This R Markdown script compares pseudosynchrony to synchrony values for the MSYNC project. The data was preprocessed before being read into this script. We use an approach described in Moulder et al. (2018, Psychol Methods) to test the following hypothesis: “Synchrony does not exist between sections of size m in these two time series” (section sliding). Then, we use the same settings as when computing synchrony to measure synchrony (windowed-lagged cross-correlation with the same timing settings).

Package versions

The following packages are used in this RMarkdown file:

```
## [1] "R version 4.5.1 (2025-06-13)"
## [1] "knitr version 1.50"
## [1] "tidyverse version 2.0.0"
## [1] "ggpubr version 0.6.0"
## [1] "BayesFactor version 0.9.12.4.7"
## [1] "effectsize version 1.0.1"
## [1] "gggrain version 0.0.4"
## [1] "rMEA version 1.2.2"
```

MEA data

```
if (!file.exists(file.path(dt.path, "mea", "df_psync.csv"))) {

  # create a dataframe
  df.psync = data.frame()

  # start with the conversation task
  attach(file.path(dt.path, "mea", "MEA_CT.Rdata"), name = 'CT')
  df.mea = pseudosync_seg(mea.scaled,
                          sampRate = 120,
                          lagSec = 5,
                          winSec = 30,
                          incSec = 15,
                          n = n)

  df.mea = df.mea %>%
    mutate(
      dyad = sprintf("MSYNC_%s", gsub("all_(.+)_0.*", "\\1", name)),
      task = "CT"
    ) %>%
    merge(., df.ccf.agg %>% ungroup() %>%
          filter(position == "B" & measure == "mean") %>%
          select(dyad, phase, MEA.sync))
  df.psync = rbind(df.psync, df.mea)
  # detach the CT data
  detach('CT')

  # at the mirror game
```

```

attach(file.path(dt.path, "mea", "MEA_MG.Rdata"), name = 'MG')
df.mea = pseudosync_seg(mea.scaled,
                        sampRate = 120,
                        lagSec = 2,
                        winSec = 30,
                        incSec = 1,
                        n = n)

df.mea = df.mea %>%
  mutate(
    dyad = sprintf("MSYNC_%s", gsub("all_(.+)_0.*", "\\1", name)),
    task = "MG",
    phase = as.numeric(substr(name, nchar(name), nchar(name)))
  ) %>%
  merge(., df.ccf.agg %>% ungroup() %>%
        filter(position == "B" & measure == "mean") %>%
        select(dyad, phase, MEA.sync))
df.psync = rbind(df.psync, df.mea)
# detach the MG data
detach('MG')

# save the pseudosynchrony values
write_csv(df.psync, file.path(dt.path, "mea", "df_psync.csv"))
} else {
  df.psync = read_csv(file.path(dt.path, "mea", "df_psync.csv"))
}

df.pagg = df.psync %>%
  group_by(dyad, task) %>%
  summarise(
    psync = mean(psync),
    MEA.sync = mean(MEA.sync)
  ) %>%
  pivot_longer(cols = c(psync, MEA.sync)) %>%
  mutate_if(is.character, as.factor)

aov.CT = anovaBF( value ~ name,
                  df.pagg %>% filter(task == "CT"))

aov.CT@bayesFactor

##              bf              error              time              code
## name -0.9078582 0.0001514629 Fri Jul  4 13:36:07 2025 40dca1a4591

interpret_bf(aov.CT@bayesFactor$bf, log = T)

## [1] "anecdotal evidence against"
## (Rules: jeffreys1961)

p1 = df.pagg %>%
  filter(task == "CT") %>%
  ggplot(., aes(x = name, y = value, fill = name, colour = name)) + #
  geom_rain(rain.side = 'f1x1',
            boxplot.args = list(color = "black", outlier.shape = NA, show_guide = FALSE, alpha = a),
            violin.args = list(color = "black", outlier.shape = NA, alpha = a),
            point.args = list(show_guide = FALSE, alpha = a)
          ) +
  stat_summary(fun = mean, geom = "line", aes(group = 1), colour = "black") +
  stat_summary(fun = mean, geom = "point", shape = 23, size = 2,
              aes(group = name)) +
  stat_summary(fun = mean, geom = "point", shape = 5, size = 2,
              aes(group = name), colour = "black") +

```

```

scale_fill_manual(values = custom.col[c(1,3)]) +
scale_colour_manual(values = custom.col[c(1,3)]) +
labs(title = "Conversation task",
      x = "",
      y = "") +
theme_bw() +
theme(legend.position = "none",
      plot.title = element_text(hjust = 0.5),
      text = element_text(size = 12))

aov.MG = anovaBF( value ~ name,
                  df.pagg %>% filter(task != "CT"))

aov.MG@bayesFactor

##           bf           error           time           code
## name 59.0867 3.015315e-31 Fri Jul 4 13:36:07 2025 40dc42cd37c5
interpret_bf(sort(aov.MG@bayesFactor$bf, decreasing = T)[1], log = T)

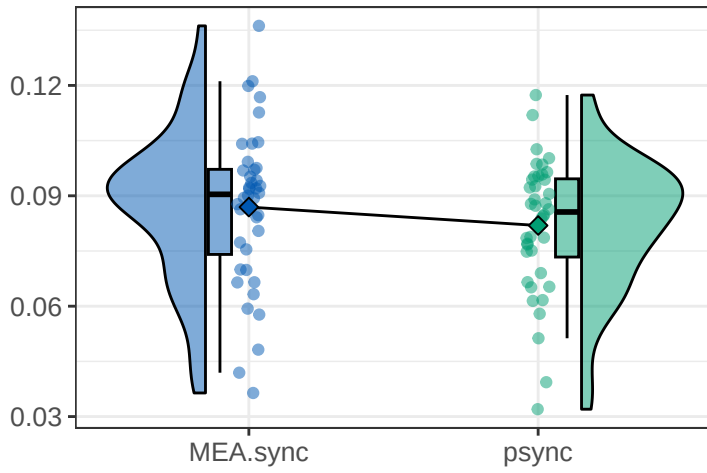
## [1] "extreme evidence in favour of"
## (Rules: jeffreys1961)

p2 = df.pagg %>%
  filter(task != "CT") %>%
  ggplot(., aes(x = name, y = value, fill = name, colour = name)) + #
  geom_rain(rain.side = 'fix1',
            boxplot.args = list(color = "black", outlier.shape = NA, show_guide = FALSE, alpha = a),
            violin.args = list(color = "black", outlier.shape = NA, alpha = a),
            point.args = list(show_guide = FALSE, alpha = a)
          ) +
  stat_summary(fun = mean, geom = "line", aes(group = 1), colour = "black") +
  stat_summary(fun = mean, geom = "point", shape = 23, size = 2,
              aes(group = name)) +
  stat_summary(fun = mean, geom = "point", shape = 5, size = 2,
              aes(group = name), colour = "black") +
  scale_fill_manual(values = custom.col[c(1,3)]) +
  scale_colour_manual(values = custom.col[c(1,3)]) +
  labs(title = "Mirror game",
        x = "",
        y = "") +
  theme_bw() +
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5),
        text = element_text(size = 12))

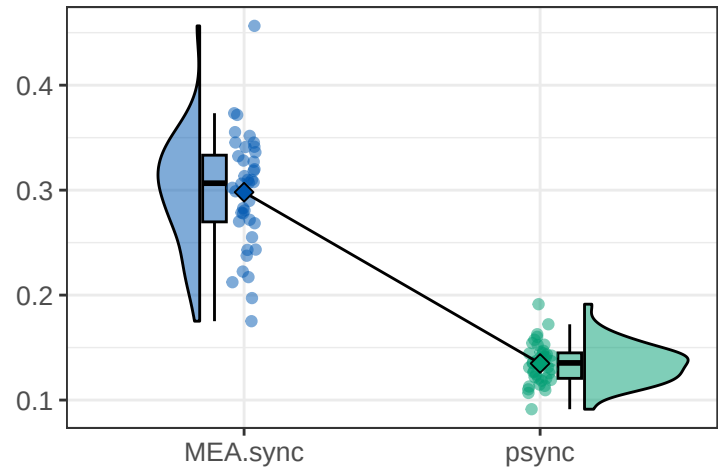
ggarrange(p1, p2, ncol = 2)

```

Conversation task



Mirror game



Synchrony in CT was not higher than pseudosync. Therefore, we will not analyse this any further for now until we figure it out. However, synchrony was higher in the MG task, so we can use these values.

OpenPose data

OpenFace data